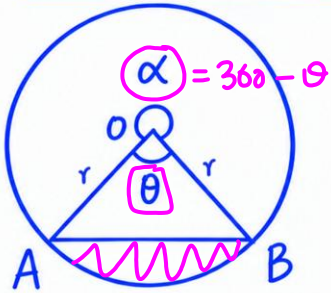


Segment of circle (All types)

Area of Segment

$\frac{\theta}{360} \times \pi r^2$

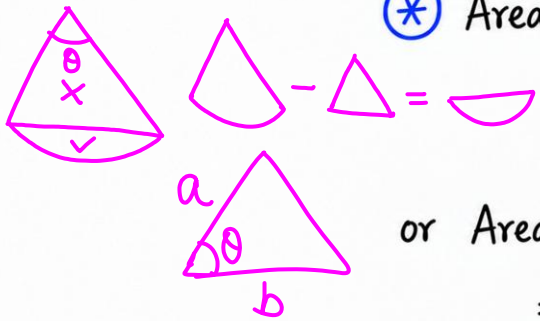


✱ Area of minor segment = minor sector AOB − △ AOB



$= \frac{\theta}{360} \times \pi r^2 - \frac{1}{2} \cdot r^2 \sin\theta$

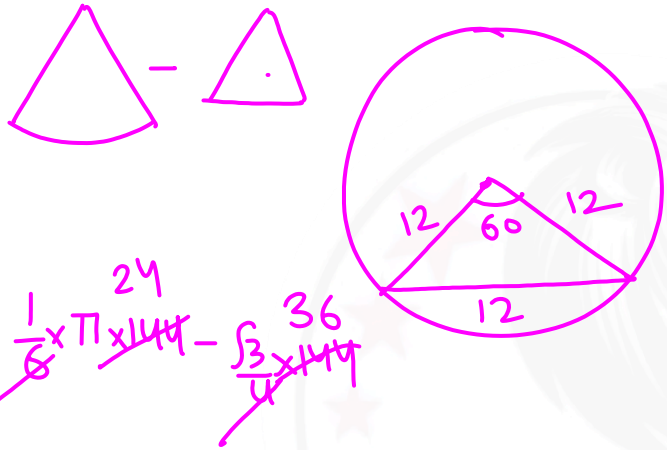
✱ Area of major segment = major sector AOB + △ AOB



$= \frac{\alpha}{360} \times \pi r^2 + \frac{1}{2} \cdot r^2 \sin\theta$

or Area of circle − Area of minor segment

$= \pi r^2 - \left(\frac{\theta}{360} \times \pi r^2 - \frac{1}{2} \cdot r^2 \sin\theta \right)$



A circle of radius of 12 cm. Two radii are drawn such that the length of the chord connecting their endpoints is 12 cm. What is the area of the minor segment formed?

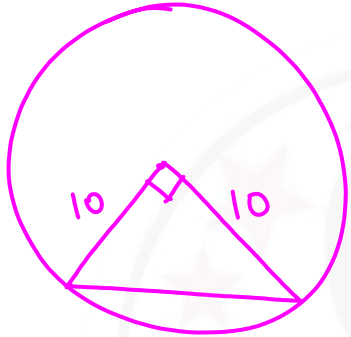
12 सेमी त्रिज्या वाले एक वृत्त में दो त्रिज्याएँ खींची जाती हैं ताकि उनके सिरों को जोड़ने वाली जीवा की लंबाई 12 सेमी हो। निर्मित छोटे खंड का क्षेत्रफल क्या है?

a) $(12\pi - 18\sqrt{3})$ sq. cm

b) $(24\pi - 36\sqrt{3})$ sq. cm

c) $(12\pi - 36\sqrt{3})$ sq. cm

d) $(24\pi - 18\sqrt{3})$ sq. cm



$\frac{1}{4} \times \frac{25}{100} \pi - \frac{1}{2} \times \frac{50}{100}$

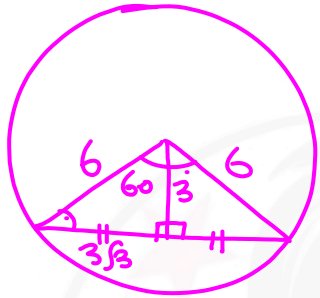
A chord in a circle with radius 10 cm subtends an angle of 90° at the center. What is the area of the minor segment?

10 सेमी त्रिज्या वाले वृत्त की एक जीवा केंद्र पर 90° का कोण बनाती है। लघु वृत्तखंड का क्षेत्रफल क्या है?

a) (25π–50) sq. cm
b) (50π–25) sq. cm
c) (50π–50) sq. cm
d) (100π–50) sq. cm

PARMAR SSC

$\frac{1}{2} \times b \times h$



What is the area of the segment formed by a chord in a circle of radius 6 cm, if the angle subtended at the center is 120°?

6 सेमी त्रिज्या वाले एक वृत्त में एक जीवा द्वारा बनाए गए खंड का क्षेत्रफल क्या है, यदि केंद्र पर अंतःस्थित कोण 120° है?

a) $12\pi - 9\sqrt{3}$

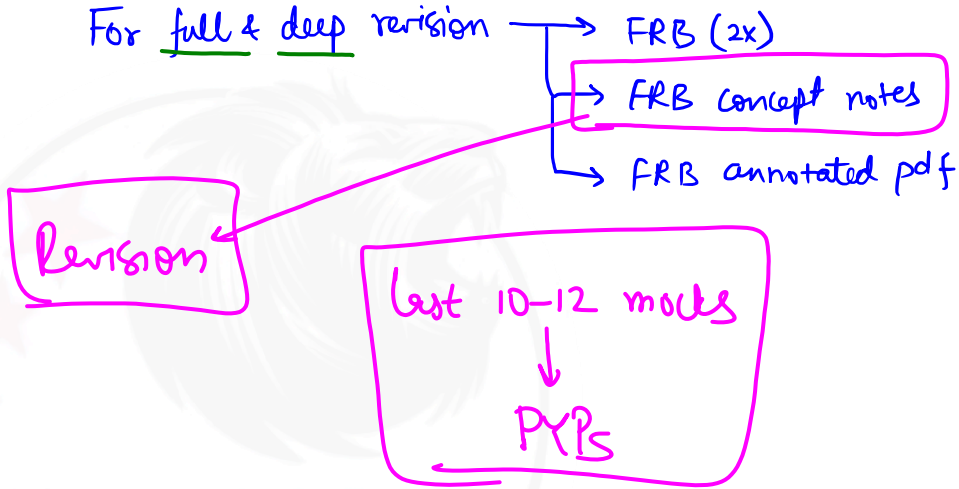
b) $12\pi - 18\sqrt{3}$

c) $36\pi - 9\sqrt{3}$

d) $12\pi - 9$

$\frac{120}{360} \times \pi \times 36 - 9\sqrt{3}$ (12)

$\frac{1}{2} \times 6 \times 6 \times \frac{\sqrt{3}}{2}$ (sin 120)


For full & deep revision
FRB (2x)
FRB concept notes
FRB annotated pdf
Revision
last 10-12 mocks
PYPs